Towards Effective Attention Approximation in Biomedical Image Segmentation: Insights from ApproxDA-TransUNet

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***Abstract*—Dual-attention mechanisms have demonstrated strong performance in medical image segmentation by jointly modeling long-range spatial and channel dependencies, but their quadratic complexity has motivated numerous approximation methods whose task-dependent behavior remains poorly understood. To bridge this gap, we propose Approximate Dual Attention TransUNet (ApproxDA-TransUNet), an efficient dual-attention approximation framework for hybrid CNN-Transformer architectures that replaces global attention with controllable windowed low-rank spatial attention and grouped channel attention. Unlike existing efficient attention methods that primarily pursue computational savings, ApproxDA-TransUNet also provides a controllable framework for systematically studying how attention approximation affects segmentation performance. Extensive experiments across five representative benchmarks (Synapse, ACDC, Kvasir-SEG, CVC-ClinicDB, and ISIC 2018) demonstrate the best performance among compared methods with consistent improvements over DA-TransUNet: 80.94%, 88.97%, 90.17%, 90.99%, and 89.58% Dice on Synapse, ACDC, Kvasir-SEG, CVC-ClinicDB, and ISIC 2018, respectively. These results also reveal two key observations: First, we identify Global Context Sensitivity (GCS): tasks requiring long-range dependencies are far more sensitive to approximation scale than texture-dominated ones, and inter-class spatial separation is the strongest observed correlate of this sensitivity among the factors we examined. Second, we observe Gate Collapse: in our evaluated configurations, the learnable attention gate settles into near-uniform routing, which a gradient-based fixed-point analysis suggests is associated with gradient symmetry between branches, undermining adaptive fusion. These findings reframe approximation as a task-dependent inductive bias: rather than a pure efficiency tool, it can improve segmentation accuracy when the structural prior induced by spatial restriction, low-rank projection, or both aligns with the task's contextual requirements. Our code and analysis offer both a strong segmentation baseline and actionable principles for designing efficient medical transformers.**

***Index Terms*—medical image segmentation, dual attention, efficient transformers, attention approximation, global context sensitivity, gate collapse**

I. Introduction

Vision Transformers (ViTs) have fundamentally transformed medical image segmentation by capturing global long-range contextual relationships that convolutional neural networks (CNNs) struggle to resolve due to their localized receptive fields [11]–[13]. Within hybrid CNN-Transformer architectures, *dual attention* [1]—which concurrently models spatial dependencies via a Position Attention Module (PAM) and cross-channel interactions via a Channel Attention Module (CAM)—has emerged as a particularly potent paradigm. Pioneering frameworks such as DA-TransUNet [2] embedded dual-attention blocks across multi-scale skip connections, establishing benchmark performance on multi-organ and polyp segmentation.

However, the computational cost of full-map dual attention scales quadratically with spatial token length ($\mathcal{O}(N^2C)$) and channel depth ($\mathcal{O}(C^2N)$). To mitigate this bottleneck, a rich body of efficient approximation strategies has emerged, including local windowed partitioning [3], low-rank matrix decomposition [4], and grouped channel factorizations [5]. A ubiquitous, yet rarely scrutinized, assumption underlying these approximation techniques is that sparse or localized attention serves purely as an engineering compromise: a computational compression step intended to approximate global interactions with negligible loss in representational fidelity.

In medical vision, however, this assumption encounters a profound challenge: biomedical segmentation tasks exhibit extreme anatomical and contextual heterogeneity. While segmenting the human pancreas in abdominal CT requires resolving spatial relationships relative to distant anatomical landmarks (such as the stomach, duodenum, and splenic vein), segmenting a localized colon polyp or cutaneous melanoma depends predominantly on fine-grained textural cues and local boundary gradients. This dichotomy raises a fundamental research question: *How does attention approximation alter representation learning across heterogeneous clinical tasks, and what intrinsic task properties govern when and at what scale approximation is beneficial?*

To address this question, we develop **ApproxDA-TransUNet**[1], a principled dual-attention framework featuring three orthogonal, independently controllable approximation axes: (i) spatial window scope (M), (ii) representation projection rank (r), and (iii) channel grouping count (G). Unlike conventional lightweight networks optimized solely for FLOP reduction, ApproxDA-TransUNet is engineered with clean modular decoupling, enabling systematic isolation of individual approximation factors without architectural con-

[1]Source code and pretrained weights will be made available upon acceptance.

founding. Furthermore, by restructuring tensor operations for contiguous memory layout, ApproxDA-TransUNet avoids a PyTorch autograd/DDP incompatibility that we encountered in our reproduced DA-TransUNet implementation, enabling native multi-GPU Distributed Data Parallel (DDP) training in our test environment.

Systematic evaluation across five diverse clinical imaging modalities (Synapse multi-organ CT, ACDC cardiac MRI, Kvasir-SEG polyp endoscopy, CVC-ClinicDB colonoscopy, and ISIC 2018 dermoscopy) reveals a structured spectrum of context sensitivity. Building on these observations, we conduct a systematic candidate-factor analysis to identify the property most consistently associated with attention window sensitivity, analyze late-stage optimization dynamics, and provide a gradient-based fixed-point analysis for multi-branch gating behavior.

The primary contributions of this paper are summarized as follows:

- **Modular ApproxDA-TransUNet Architecture with DDP-Compatible Implementation:** We propose a modular dual-attention framework with three independently controllable approximation axes (M, r, G) that consistently matches or outperforms dense full-attention baselines across five clinical benchmarks, and which avoided a DDP incompatibility we encountered with our reproduced DA-TransUNet baseline in our test environment.
- **Discovery of GCS and the SSD Hypothesis:** Across five diverse imaging modalities, we observe a $4.6\times$ difference in single-run window-sensitivity spans (ΔDSC span: $0.50\,\text{pp}$ to $2.30\,\text{pp}$), formalizing this phenomenon as *Global Context Sensitivity (GCS)*. Using ACDC as a key discriminating benchmark, we evaluate five candidate spatial metrics and find that inter-class window separation (SC5, $\rho = +\,0.89$) is the candidate most consistently associated with GCS across all evaluated benchmarks—an interpretive finding we term *Semantic Spatial Dependency (SSD)*.
- **Optimization Dynamics & Gate Collapse Analysis:** We observe that localized windowing is associated with reduced late-stage validation variability on low-GCS tasks (standard deviation reduced by up to $18.2\times$). We further provide a gradient-based fixed-point analysis showing that symmetric dual-branch architectures tend toward uniform gating ($g \approx 0.5$), supported by empirical gate measurements across training.
- **Exploratory Design Heuristics:** We discuss practical heuristics for attention architecture design based on our findings, including pre-training SC5 assessment as a lightweight indicator of expected window sensitivity.

The remainder of this paper is organized as follows: Section II reviews related literature; Section III presents the ApproxDA-TransUNet framework and complexity formulations; Section IV reports results against state-of-the-art baselines across five benchmarks; Section V investigates the GCS spectrum and evaluates candidate mechanisms including Semantic Spatial Dependency (SSD); Section VI analyzes training optimization volatility and gate collapse dynamics; Section VII concludes with practical design guidelines.

II. Related Work

A. CNN-based Medical Image Segmentation

Convolutional encoder-decoder architectures established the foundational paradigm for medical image segmentation, with U-Net [6] introducing the skip-connection design that underpins most segmentation networks, and successors [7]–[10] progressively adding dense connections, attention gating, and residual learning for improved feature reuse and training stability. These models offer strong computational efficiency and favorable inductive biases for local texture patterns, but their fixed receptive fields cannot capture long-range spatial dependencies across anatomical structures, limiting performance on tasks requiring global context.

B. Transformer-based Medical Image Segmentation

Transformer architectures have significantly advanced medical image segmentation by enabling global dependency modeling beyond the locality of convolutional neural networks. Representative methods such as TransUNet [11], Swin-Unet [12], UNETR [13], UCTransNet [14], TransNorm [15], and DAE-former [16] combine convolutional feature extraction with transformer-based contextual modeling to achieve state-of-the-art segmentation performance across a wide range of medical imaging tasks. These methods treat global context modeling as uniformly beneficial, however, without examining whether different tasks require different degrees of spatial context.

C. Dual Attention and DA-TransUNet

DANet [1] introduced Position Attention Modules (PAM) and Channel Attention Modules (CAM), capturing spatial and channel correlations at $\mathcal{O}(N^2C)$ and $\mathcal{O}(C^2N)$ complexity respectively. DA-TransUNet [2] adapted this design for medical image segmentation by inserting dual-attention blocks at multiple skip connections of a hybrid R50-ViT-B/16 encoder-decoder, achieving state-of-the-art accuracy while inheriting the quadratic cost of full-map attention. DBAANet [17] further extends dual-branch attention by aggregating position and channel attention across multiple scales, demonstrating the continued appeal of dual-attention paradigms for medical image segmentation.

D. Efficient Attention and Adaptive Routing

Windowed local attention [3], low-rank projection [4], and grouped channel interaction [5] have become standard strategies for reducing the cost of global attention, with learnable gating widely adopted to fuse multi-branch outputs [5], [16]. DAMamba [18] further shows that dual-attention routing transfers effectively to state-space models for lightweight breast ultrasound segmentation, confirming the versatility of multi-branch attention beyond standard transformer architectures. Existing efficient attention methods primarily optimize computational efficiency while implicitly assuming approximation is universally applicable. Whether approximation itself changes representation learning, and why it behaves differently across

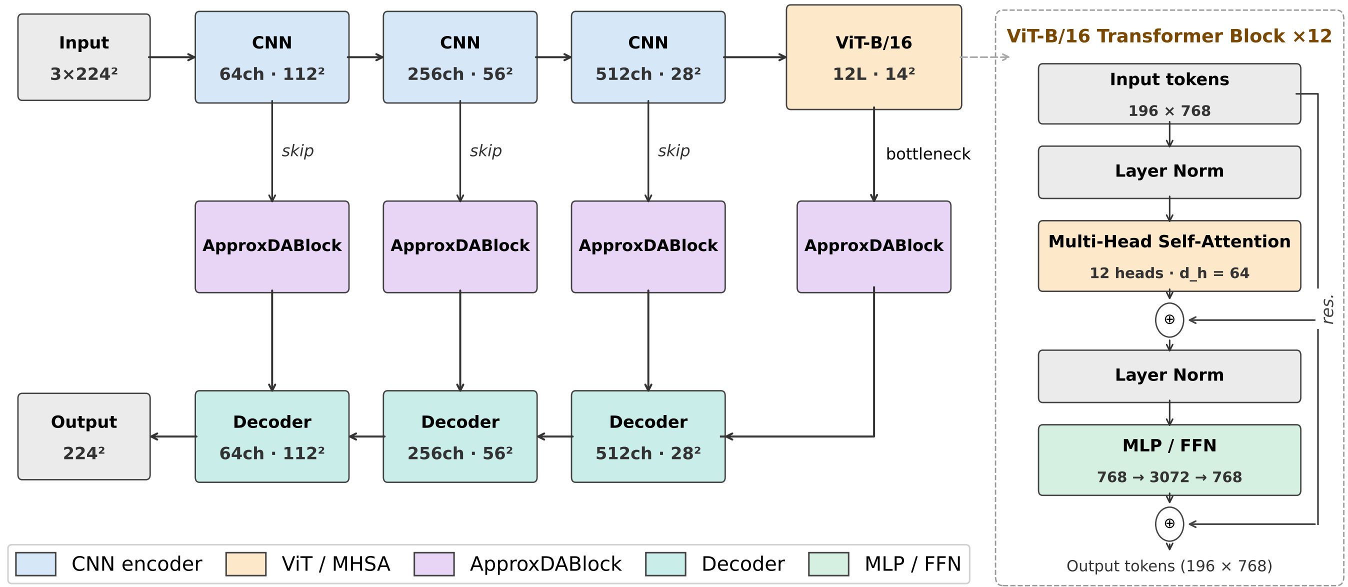

Fig. 1. ApproxDA-TransUNet overall architecture. ApproxDABlocks replace standard dual-attention modules across all skip connections and the bottleneck.

tasks with different contextual requirements, remains largely unexplored. This work addresses both questions by studying the context-sensitive behavior of approximation across heterogeneous segmentation tasks and exposing the gradient symmetry dynamics that cause adaptive routing to degenerate into fixed blending.

III. Method

This section introduces the overall architecture of ApproxDA-TransUNet and the placement of ApproxDABlocks within it, then details the three independently controllable approximation axes each block implements, and finally analyzes their computational complexity.

A. Architecture Overview

Figure 1 illustrates the overall architecture: ApproxDA-TransUNet adopts the hybrid CNN–ViT encoder-decoder structure of DA-TransUNet [2], with an R50-ViT-B/16 backbone pretrained on ImageNet-21K. Three CNN encoder stages produce multi-scale features at strides 4, 8, and 16; a 12-layer ViT processes 14×14 patch tokens. ApproxDABlocks replace all dual-attention modules at three skip connections—512 channels at 28×28, 256 at 56×56, and 64 at 112×112—and at the bottleneck. Each block routes activations through a Low-Rank Windowed PAM branch and a Grouped CAM branch, fused by a configurable gate g. Figure 2 details the internal structure of each ApproxDABlock and its three independently controllable approximation axes.

B. Attention Approximation Formulation

The original DA module performs dense spatial and channel attention over the full feature map, with quadratic complexity in both spatial resolution and channel dimension. ApproxDA-TransUNet reduces this cost along three independent axes, each targeting a distinct aspect of the approximation:

Spatial scope (window size M) controls *how far* PAM reaches. Full global attention ($M{=}H$) captures all long-range spatial relationships; restricting to $M\times M$ local windows trades long-range context for lower complexity. This axis most directly governs whether tasks with broader contextual requirements retain the anatomical context they depend on.

Representation fidelity (rank r) controls *how precisely* attention is computed within each window. Low-rank projection ($r \ll M^2$) approximates the affinity matrix at reduced cost; higher r approaches exact attention.

Channel interaction (groups G) controls *which channels* may interact in CAM. Partitioning C channels into G independent groups limits cross-channel dependency to within-group interactions, reducing the interaction scope.

The axes are architecturally independent: any two of $\{M, r, G\}$ can be held fixed while the third varies. In this paper, M is the axis we systematically ablate (Section V-A); gate routing is studied as a separate mechanism (Section VI-B); rank r and groups G are exposed as configurable capabilities of the framework, with their held-fixed default values ($r{=}32$, $G{=}8$) used throughout the main results.

C. Spatial Approximation

Instead of computing global position attention, ApproxDA-TransUNet partitions the feature map into non-overlapping windows of size $M\times M$. Attention is computed independently inside each local window, reducing complexity from $\mathcal{O}(N^2C)$ to $\mathcal{O}(NM^2C)$.

Window partitioning. Following [3], we partition the feature map into n_W non-overlapping $M\times M$ windows. Window

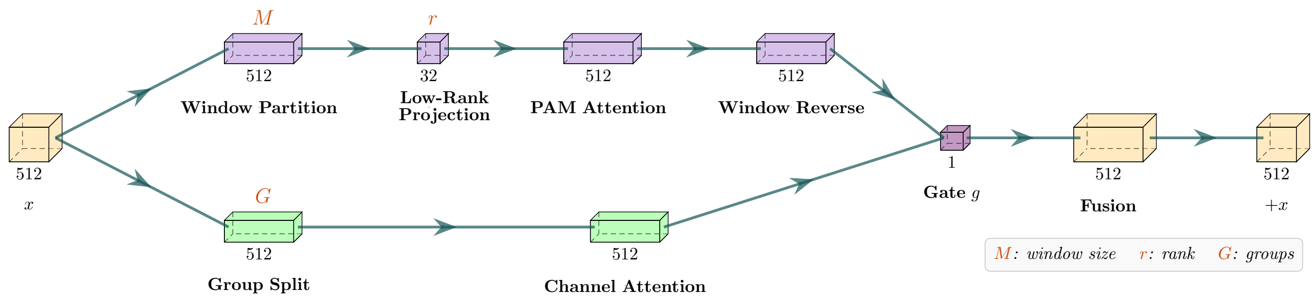

Fig. 2. ApproxDABlock internal structure with three independent approximation axes: spatial scope (M), projection rank (r), and channel grouping (G). Input $\mathbf{x}$ splits into a Low-Rank Windowed PAM branch and a Grouped CAM branch fused by gate g.

clamping $M \leftarrow \min(M, H, W)$ allows $M{=}112$ to yield global attention at all scales, enabling a clean ablation that isolates windowing from low-rank as the performance bottleneck.

Low-rank projection. Following [4], we compute query, key, and value feature matrices within each window via $1{\times}1$ convolutions, $\mathbf{Q}, \mathbf{K}, \mathbf{V} \in \mathbb{R}^{N_w \times C}$ where $N_w{=}M^2$. A shared learned projection $\mathbf{E} \in \mathbb{R}^{r \times N_w}$ ($r \ll N_w$) reduces the sequence length of the key and value:

$$\mathbf{K}' = \mathbf{E}\mathbf{K}, \quad \mathbf{V}' = \mathbf{E}\mathbf{V} \in \mathbb{R}^{r \times C}. \tag{1}$$

Attention logits are computed between the full-length query and the rank-r projected key, normalized, and applied to the projected value:

$$\mathrm{Attn}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \mathrm{softmax}\big(\mathbf{Q}\mathbf{K}'^{\top}\big)\, \mathbf{V}' \in \mathbb{R}^{N_w \times C}, \tag{2}$$

followed by a learnable residual scale, $\mathbf{y} = \alpha \cdot \mathrm{Attn}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) + \mathbf{x}$, with scalar α initialized to zero (matching our implementation, no $1/\sqrt{C}$ temperature scaling is applied to the logits). Projecting $\mathbf{K}$ and $\mathbf{V}$ costs $\mathcal{O}(N_w r C)$; computing $\mathbf{Q}\mathbf{K}'^{\top}$ and its product with $\mathbf{V}'$ each cost $\mathcal{O}(N_w r C)$, avoiding the $\mathcal{O}(N_w^2 C)$ cost of full in-window attention. Summed over N/N_w windows, total attention complexity is $\mathcal{O}(NrC)$. Combined, windowing and low-rank projection reduce the theoretical attention-matrix size from $\mathcal{O}(N^2)$ to $\mathcal{O}(Nr)$, a factor of N/r ($\sim$390$\times$ at 112$\times$112 with $r{=}32$); note this is a memory/FLOP-count reduction in the attention operation specifically; end-to-end model GFLOPs also include the projection cost $\mathcal{O}(N_w r C)$ and the surrounding convolutional layers, which do not shrink at the same rate (Table VI). Window size M forms the primary experimental axis investigated in this study, while rank $r{=}32$ is held fixed in the clean window ablations (Section V-A).

D. Channel Approximation

The original Channel Attention Module computes a $C{\times}C$ channel attention matrix at $\mathcal{O}(C^2 N)$ cost. ApproxDA-TransUNet approximates this by partitioning C channels into G groups of width $C_g = C/G$, computing independent per-group attention:

$$X_{ji}^{(g)} = \frac{\exp(\mathbf{A}_i^{(g)} \cdot \mathbf{A}_j^{(g)})}{\sum_{k=1}^{C_g} \exp(\mathbf{A}_k^{(g)} \cdot \mathbf{A}_j^{(g)})}, \tag{3}$$

with total cost $\mathcal{O}(C^2 N/G)$. Compared with dense channel interaction, grouped attention significantly reduces computational complexity while preserving local channel dependencies.

The grouping number G provides another controllable approximation dimension, allowing the influence of channel approximation to be studied independently from spatial approximation.

E. Routing Strategy

The routing of DA-TransUNet [2] is extended to a configurable `gate_mode`. PAM and CAM outputs are fused via:

$$\mathbf{y} = \mathbf{W}_f\Big(g \cdot \hat{\mathbf{P}} + (1-g) \cdot \hat{\mathbf{C}}\Big) + \mathbf{x}, \tag{4}$$

where $\hat{\mathbf{P}}$ and $\hat{\mathbf{C}}$ are the normalized PAM and CAM outputs, $\mathbf{W}_f$ is a $1{\times}1$ convolution, and g is determined by the `gate_mode` parameter:

`pam`:	$g = 1.0$	(PAM only)
`cam`:	$g = 0.0$	(CAM only)
`fixed`:	$g = 0.5$	(equal blend)
`learn`:	$g = \sigma(\mathbf{W}_g \cdot \mathrm{GAP}(\mathbf{x})) \in (0,1)^C$	

In gate=learn, $g \in (0,1)^C$ is a per-channel routing coefficient broadcast over all spatial positions; it collapses to a channel-wise blend rather than a spatially-adaptive or input-dependent scalar. This single-parameter design enables systematic ablation of routing strategy without any other architectural change. ApproxDA-TransUNet does not treat this routing module as a primary architectural contribution; rather, it serves as an analyzable component whose optimization dynamics—including an unexpected convergence to $g{\approx}0.5$ regardless of window size—are investigated in Section VI-B.

The network is trained with a combined Dice and cross-entropy objective with equal weights:

$$\mathcal{L} = \tfrac{1}{2}\mathcal{L}_{\text{Dice}} + \tfrac{1}{2}\mathcal{L}_{\text{CE}}, \tag{5}$$

following DA-TransUNet [2] for direct comparability. Dice loss optimizes the overlap metric used at evaluation and handles class imbalance naturally; cross-entropy provides stable per-pixel gradient signal throughout training.

IV. Results

We evaluate ApproxDA-TransUNet on five benchmarks spanning tasks with fundamentally different spatial context requirements, comparing against DA-TransUNet and other competitive baselines.

A. Experimental Setup

Datasets. We evaluate on five representative medical image segmentation benchmarks: Synapse multi-organ CT [19] (8 organ classes; mean DSC and HD95), ACDC cardiac MRI [24] (3 cardiac structures; DSC and HD95), Kvasir-SEG [20] (1,000 endoscopy polyp images; DSC and mIoU), CVC-ClinicDB [21] (612 colonoscopy polyp images; DSC, mIoU, and HD95), and ISIC 2018 [22] (2,594 dermoscopy skin lesion images; DSC and mIoU).

Implementation Details. All experiments use an R50-ViT-B/16 backbone pretrained on ImageNet-21K, trained on NVIDIA Tesla T4 (16 GB) GPUs. Optimizer: SGD, lr=0.01; batch size 24; input 224×224; 300 epochs. Best checkpoint and all hyperparameters (window size M, gate mode, rank r) are selected based on validation DSC evaluated every 15 epochs on the held-out validation split; the test set is used only for final reporting. Default ApproxDA-TransUNet configuration: M=7, r=32, G=8.

Gate Protocol. ApproxDA-TransUNet supports four gate modes: `pam` (g=1.0, PAM only), `cam` (g=0.0, CAM only), `fixed` (g=0.5, equal blend), and `learn` (g learned per channel). Each experiment's gate configuration is specified in the corresponding table footnote.

B. Comparison with State-of-the-Art

Tables I–V compare ApproxDA-TransUNet against published baselines across all five benchmarks.

Synapse. Table I benchmarks against 11 published methods. CNN-based models plateau at 76–78% DSC, with Att-UNet leading at 77.77%. Transformer-based methods progressively improve: TransUNet (77.48%), UCTransNet (78.23%), MIM (78.59%), Swin-Unet (79.13%), and DA-TransUNet (79.80%). ApproxDA-TransUNet with gate=pam, M=28, r=32 achieves **80.94%** DSC and 27.49 mm HD95—the highest DSC across all compared methods, surpassing DA-TransUNet by +1.14%. The intermediate window M=28 captures sufficient cross-organ context while limiting spurious long-range correlations; Section V analyzes this behavior in detail.

ACDC. On cardiac MRI segmentation (Table II), DA-TransUNet achieves a mean DSC of 88.50% (RV 88.84%, Myo **85.98%**, LV 90.67%). ApproxDA-TransUNet with gate=pam, M=7 achieves a mean DSC of 88.97% (+0.47% over DA-TransUNet), attaining the highest per-class DSC among compared methods on RV (**89.61%**) and remaining competitive on Myo (<u>85.91%</u>, essentially on par with DA-TransUNet's 85.98%). Although ApproxDA improves LV over DA-TransUNet (+0.73%, 91.40% vs. 90.67%), it remains below the global Transformer baselines on this structure (TransUNet 95.73%, Swin-Unet 95.83%). The RV result is consistent with the inductive-bias interpretation—local windowed attention benefiting a locally-resolvable, crescent-shaped structure—while Myo shows no clear windowing benefit or penalty, and LV cavity delineation, which requires estimating global cavity extent spanning multiple M=7 windows, is where both dual-attention architectures trail global-attention Transformers. Within a nominally low-GCS dataset (ΔDSC=0.73 pp overall), ApproxDA maintains an advantage over full-attention DA-TransUNet on 2 of 3 structures (RV, LV) and mean DSC.

TABLE I Segmentation Results on Synapse Multi-Organ CT

Method	DSC (%) ↑	HD95 (mm) ↓
CNN-based		
U-Net [6]	76.85	39.70
U-Net++ [7]	76.91	36.93
Residual U-Net [9]	76.95	38.44
MultiResUNet [10]	77.42	36.84
Att-Unet [8]	77.77	36.02
Transformer-based		
TransUNet [11]	77.48	31.69
UCTransNet [14]	78.23	26.75
TransNorm [15]	78.40	30.25
MIM [23]	78.59	26.59
Swin-Unet [12]	79.13	**21.55**
DA-TransUNet [2]	<u>79.80</u>	<u>23.48</u>
Dual-attention approximation (ours)		
ApproxDA-TransUNet (ours)	**80.94**	27.49

ApproxDA: gate=pam, M=28, r=32, G=8.

TABLE II Segmentation Results on ACDC Cardiac MRI

Method	RV	Myo	LV	Mean (%) ↑
U-Net [6]†	87.10	80.63	94.92	87.55
Att-UNet [8]†	87.58	79.20	93.47	86.75
TransUNet [11]	<u>88.86</u>	84.53	<u>95.73</u>	<u>89.71</u>
Swin-Unet [12]	88.55	85.62	**95.83**	**90.00**
DA-TransUNet [2]‡	88.84	**85.98**	90.67	88.50
ApproxDA-TransUNet (ours)	**89.61**	<u>85.91</u>	91.40	88.97

†R50-backbone variant reported in [11]. ‡Re-run under our protocol. ApproxDA: gate=pam, M=7, r=32, G=8.

Kvasir-SEG. On polyp segmentation (Table III), U-Net leads the CNN-based group (88.22% DSC) while TransUNet (87.91%) and DA-TransUNet† (88.44%) are the strongest Transformer-based entries. ApproxDA-TransUNet with gate=pam, M=56 achieves **90.17%** DSC, **84.27%** mIoU, and **44.35 mm** HD95—the best among compared methods across all three metrics, with a +1.73% DSC gain and −8.69 mm HD95 improvement over the nearest baseline.

CVC-ClinicDB. On colonoscopy polyp segmentation (Table IV), DA-TransUNet achieves 89.47% DSC and 82.51% mIoU—the strongest published baseline on this dataset. ApproxDA-TransUNet with gate=pam, M=112 achieves **90.99%** DSC, **85.09%** mIoU, and **14.77 mm** HD95—surpassing the nearest baseline by +1.52% DSC and +2.58% mIoU. The DSC span across $M\in\{7, 28, 56, 112\}$ is 0.62 pp,

TABLE III Segmentation Results on Kvasir-SEG Polyp

Method	DSC (%) ↑	mIoU (%) ↑	HD95 (mm) ↓
U-Net [6]	88.22	80.12	–
Att-Unet [8]	86.61	78.01	–
U-Net++ [7]	86.57	77.67	–
Res-UNet [9]	77.85	66.04	–
TransUNet [11]	87.91	80.03	–
DA-TransUNet [2]†	<u>88.44</u>	<u>81.70</u>	<u>53.04</u>
ApproxDA-TransUNet (ours)	**90.17**	**84.27**	**44.35**

†Re-run under our protocol (88.47% reported). ApproxDA: gate=pam, M=56, r=32.

confirming CVC-ClinicDB is window-robust—consistent with its binary task structure and low GCS, as analyzed in Section V.

TABLE IV Segmentation Results on CVC-ClinicDB Polyp

Method	DSC (%) ↑	mIoU (%) ↑	HD95 (mm) ↓
U-Net [6]	86.93	78.21	–
Att-Unet [8]	87.41	79.35	–
U-Net++ [7]	87.14	78.47	–
Res-UNet [9]	74.22	59.02	–
TransUNet [11]	89.01	81.63	–
DA-TransUNet [2]	<u>89.47</u>	<u>82.51</u>	–
ApproxDA-TransUNet (ours)	**90.99**	**85.09**	**14.77**

ApproxDA: gate=pam, M=112, r=32.

ISIC 2018. On skin lesion segmentation (Table V), CNN-based methods cluster between 83–88% DSC, with TransUNet (88.78%) and DA-TransUNet (88.88% reported) leading among Transformer-based approaches. ApproxDA-TransUNet with gate=learn, M=7 achieves **89.58%** DSC and 82.66% mIoU, attaining the best DSC across all compared methods: +0.70% over the originally reported DA-TransUNet value, and +1.15% over our controlled re-run of DA-TransUNet under identical protocol (88.43%, Table XI). This window-robust pattern is consistent with the low-GCS behavior observed on Kvasir-SEG (GCS defined in Section V-A).

TABLE V Segmentation Results on ISIC 2018 Skin Lesion

Method	DSC (%) ↑	mIoU (%) ↑
U-Net [6]	87.22	81.14
Att-Unet [8]	87.60	81.51
U-Net++ [7]	87.30	81.33
Res-UNet [9]	83.32	76.51
TransUNet [11]	88.78	82.63
DA-TransUNet [2]†	<u>88.88</u>	**82.78**
ApproxDA-TransUNet (ours)	**89.58**	<u>82.66</u>

†Originally reported value; our controlled re-run under identical protocol gives 88.43% DSC (Table XI). ApproxDA: gate=learn, M=7, r=32, G=8.

Qualitative Analysis. Across five benchmarks (Figure 3), ApproxDA-TransUNet achieves the best or competitive results among compared methods. The qualitative comparisons illustrate clear behavioral distinctions among the evaluated architectures:

- **U-Net** [6], as a pure CNN baseline lacking long-range dependency modeling, often produces boundary leakage into ambiguous background regions (e.g., lower-left false positive in Kvasir-SEG) and irregular, jagged contours on skin lesions in ISIC 2018.
- **TransUNet** [11] mitigates false-positive detections by incorporating ViT self-attention into the bottleneck to encode global spatial context. However, lacking fine-grained spatial-channel dual attention or localized boundary refinement, its predictions tend to be overly smooth, missing subtle lesion margins.
- **DA-TransUNet** [2] captures dense spatial and channel dependencies via full PAM and CAM, improving upon TransUNet. Nonetheless, unconstrained global spatial attention can introduce subtle boundary blurriness on low-GCS tasks where distant pixel interactions add non-informative contextual noise.
- **ApproxDA-TransUNet (ours)** delivers superior boundary conformity and suppresses spurious background artifacts across datasets (achieving slice DSCs of 0.971 on Kvasir-SEG and 0.975 on ISIC 2018). The restricted spatial attention window preserves fine-grained texture priors, providing a beneficial locality prior in the shown Kvasir-SEG and ISIC 2018 examples.

C. Efficiency Analysis

Although the proposed approximation substantially reduces the theoretical complexity of the dual-attention module, the overall computational cost is dominated by the ViT backbone and convolutional encoder, which remain unchanged. As shown in Table VI, end-to-end GFLOPs and memory exhibit only marginal differences from DA-TransUNet. The practical advantage of ApproxDA-TransUNet therefore lies in improved architectural compatibility rather than total model complexity reduction: in our test environment, our reproduced DA-TransUNet implementation encountered a PyTorch autograd/DDP incompatibility that prevented multi-GPU training, while ApproxDA-TransUNet trains natively across multiple GPUs, enabling native DDP training and reducing wall-clock training time in our two-GPU setup (8.3h vs. 11.4h; note this reflects reduced wall-clock time, not reduced total GPU-hours, since training uses 2 GPUs concurrently).

D. Summary and Transition to Mechanistic Analysis

The quantitative and qualitative evaluations across five heterogeneous benchmarks highlight two core empirical observations:

1) **Task-aligned approximation consistently outperforms full dual attention across benchmarks:** Across all five benchmarks, ApproxDA-TransUNet surpasses DA-TransUNet (+1.14% on Synapse, +0.47% on ACDC, +1.73% on Kvasir-SEG, +1.52% on CVC-ClinicDB, and +0.70%/+1.15% on ISIC 2018 vs. reported/re-run baseline respectively), suggesting that dense full dual attention is not always optimal and may introduce non-informative interactions on some tasks. Note that the best-performing configuration is

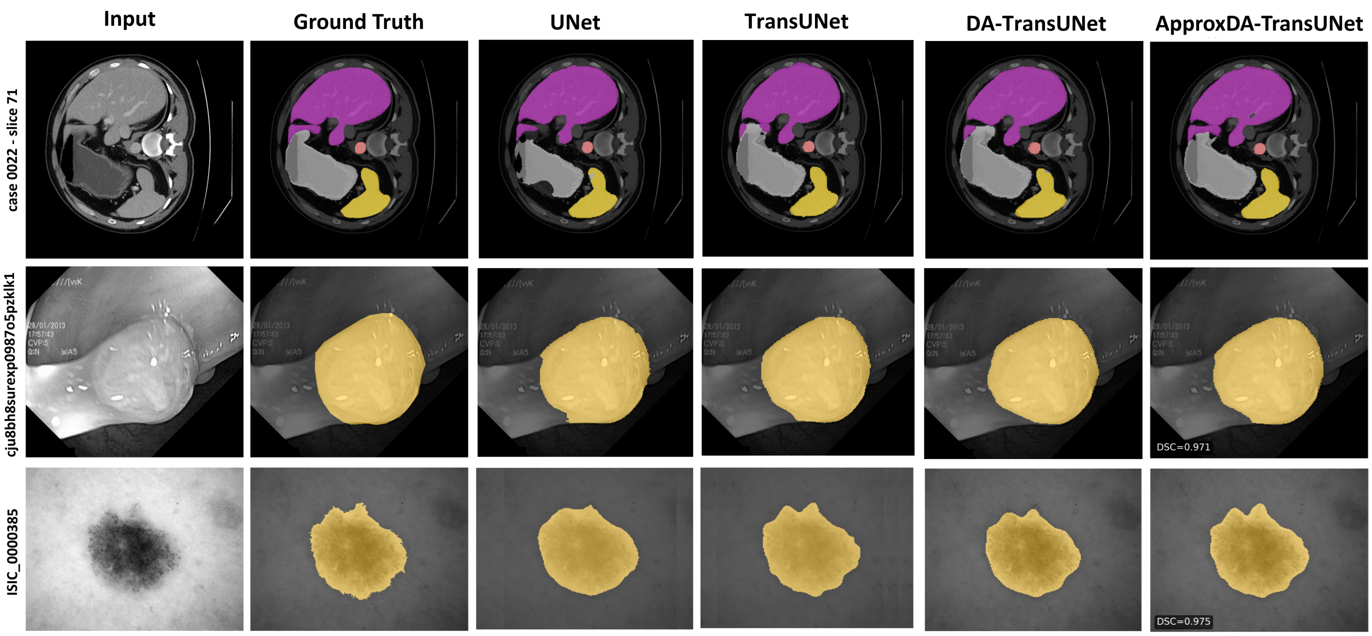

Fig. 3. Qualitative segmentation comparisons across representative benchmarks: Synapse multi-organ CT (M=28, top), Kvasir-SEG polyp (M=56, middle), and ISIC 2018 skin lesion (M=7, bottom). ApproxDA-TransUNet produces sharper organ/lesion contours while effectively suppressing false-positive boundary leakage.

TABLE VI EFFICIENCY COMPARISON ON SYNAPSE (NVIDIA TESLA T4)

Method	Params (M)	GFLOPs	Train VRAM	Train Time	Infer (s/vol)	Multi-GPU
DA-TransUNet [2]	107.95	30.2	11.5 G (1 GPU)	11.4h	120.0	No†
ApproxDA-TransUNet (gate=pam, M=7)	112.98	32.1	6.4 G/GPU (×2 GPU, total 12.8 G)‡	8.3h (×2 GPU)	121.3	Yes (DDP)
ApproxDA-TransUNet (gate=learn, M=7)	114.90	32.1	10.6 G (1 GPU)	11.5h	114.2	Yes (DDP)

†In our test environment, the reproduced DA-TransUNet implementation encountered an implementation-specific PyTorch autograd/DDP incompatibility that prevented multi-GPU training; memory is reported on a single GPU. ‡Per-GPU memory with DDP batch splitting across 2 GPUs; total memory footprint is 12.8 G. Memory comparison between single-GPU and multi-GPU configurations should be interpreted with this difference in mind.

not uniformly the most local one: CVC-ClinicDB's best result uses M=112 (global spatial scope with low-rank approximation only) and Kvasir-SEG's uses M=56, so the benefit reflects task-aligned approximation broadly (spatial restriction, low-rank projection, or both) rather than compact locality specifically.

2) **Spatial context sensitivity is intrinsically task-dependent:** While multi-organ CT (Synapse) demands careful window tuning to balance local detail and cross-organ context, cardiac, polyp, and skin lesion tasks exhibit broad performance plateaus.

These findings challenge the conventional view of attention approximation as a mere efficiency compromise and motivate two fundamental questions: *What intrinsic task properties dictate this context sensitivity (investigated in Section V)?* and *How does spatial approximation alter training dynamics and architectural routing (investigated in Section VI)?*

V. GLOBAL CONTEXT SENSITIVITY AND CANDIDATE MECHANISMS

Section IV demonstrated that the impact of attention approximation is deeply task-dependent. This section investigates the theoretical and empirical underpinnings of this phenomenon. We establish the empirical spectrum of Global Context Sensitivity (GCS), systematically evaluate alternative geometric and modality-based candidate explanations, and identify Semantic Spatial Dependency (SSD) as the candidate most consistently associated with GCS across the evaluated benchmarks.

A. Empirical Window Sensitivity and the GCS Spectrum

We first examine how spatial approximation scale (window size M) impacts segmentation accuracy across different biomedical imaging tasks. Table VII isolates this effect on Synapse multi-organ CT by varying $M \in \{7, 28, 56, 112\}$ under fixed gate=pam, r=32. A separate gate-collapse robustness check at M=14, r=64, gate=learn is reported in Table XII (Section VI-B), not here, since it varies rank and gate simultaneously with window size and cannot isolate the window-size effect.

The Synapse DSC-vs-M curve exhibits a clear non-monotonic profile: performance peaks at M=28 (80.94% DSC, +1.14% over DA-TransUNet), while overly local (M=7: 78.64%) and overly global (M=112: 79.44%) scopes underperform. This behavior mirrors a bias-variance tradeoff: insufficient receptive fields miss critical cross-organ context,

TABLE VII Spatial Approximation Ablation on Synapse (gate=pam, r=32, G=8)

Config	M	r	Gate	DSC (%) ↑	HD95 (mm) ↓
Windowed + low-rank	7	32	pam	78.64	31.09
Windowed + low-rank	28	32	pam	**80.94**	<u>27.49</u>
Windowed + low-rank	56	32	pam	78.90	35.23
Global + low-rank	112	32	pam	<u>79.44</u>	**27.26**

M=112 via window clamping yields full-map attention at all scales.

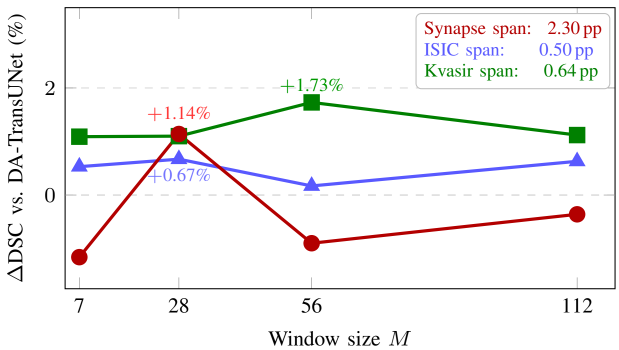

Fig. 4. ΔDSC vs. window size M (gate=pam, r=32) across representative tasks spanning the GCS spectrum. Curve colors: Synapse (red), ISIC 2018 (blue), Kvasir-SEG (green).

whereas unrestricted global attention admits non-informative long-range correlations from irrelevant background tissue.

To quantify task sensitivity across benchmarks, we define empirical **Global Context Sensitivity (GCS)** as the maximum performance span across window scales:

$$\Delta\text{DSC} = \max_{M} \text{DSC}(M) - \min_{M} \text{DSC}(M), \tag{6}$$

measured under gate=pam, r=32, from single training runs per configuration. As summarized in Table VIII and Figure 4, tasks form a structured spectrum spanning an observed 4.6× difference in single-run window-sensitivity spans:

- **High-GCS Tier (Synapse, ΔDSC = 2.30 pp):** Steep performance curve with a sharp optimum at M=28. Window choice is critical to accuracy.
- **Low-GCS Tier (ACDC, Kvasir, CVC, ISIC, ΔDSC ≤ 0.73 pp):** Broad performance plateaus where any local window ($M \in \{7, 28, 56\}$) achieves competitive or superior performance relative to global attention.

TABLE VIII Empirical GCS Spectrum (ΔDSC) Across Five Biomedical Segmentation Benchmarks (gate=pam, r=32).

Dataset	Modality	Fg. Cls.	ΔDSC (pp)	Tier
Synapse	Multi-organ CT	8	**2.30**	High
ACDC	Cardiac MRI	3	0.73	Low
Kvasir-SEG	Polyp (endo.)	1	0.64	Low
CVC-ClinicDB	Polyp (colono.)	1	0.62	Low
ISIC 2018	Skin lesion	1	0.50	Low

Fg. Cls.: number of foreground semantic classes (excluding background), matching Section IV. Distinct from the mean observed per-slice/per-image foreground class count used as a proxy in Table IX.

B. Candidate-Factor Analysis: Identifying the Strongest GCS Correlate

What intrinsic property of a medical segmentation task is most consistently associated with its GCS tier? We conduct a systematic candidate-factor analysis using training-free annotation mask statistics computed across 200 randomly sampled masks per dataset. We formulate and evaluate five candidate hypotheses:

- **SC1 — Foreground Area Ratio:** Fraction of foreground pixels. Tests if target scale dictates window sensitivity.
- **SC2 — Boundary Complexity:** Mean isoperimetric ratio ($P^2/4\pi A$) per component. Tests if boundary irregularity drives GCS.
- **SC3 — Spatial Entropy:** Normalized foreground entropy over an 8×8 grid. Tests if target dispersion determines context requirements.
- **SC4 — Window Crossing Ratio (WCR):** Fraction of foreground pixels belonging to connected components spanning more than one $M \times M$ window (M=28 px in pixel space). Tests if component-level window boundary crossing drives sensitivity.
- **SC5 — Inter-Class Window Separation:** Fraction of semantic class-centroid pairs falling in distinct $M \times M$ windows. Computed on annotation masks using M=28 px, which corresponds to the M=7 feature-space window at the highest-resolution skip stage under the standard 4× encoder downsampling of Synapse images (224 px input); other skip stages and the bottleneck operate at coarser downsampling and are not directly represented by this pixel-space value. For single-foreground-class (binary) datasets, no inter-class pair exists, so SC5 is undefined in the literal sense; we set it to zero by convention to encode the absence of inter-class separation. This convention does not imply the absence of within-class long-range or topological dependencies, and SC5 is accordingly not diagnostic of within-class global structure in binary tasks. Measures whether multi-class reasoning requires cross-window communication.

A candidate consistent with GCS should achieve strong positive rank correlation ($\rho > 0.8$) with the empirical GCS ranking across all five datasets (Synapse > ACDC > Kvasir ≈ CVC > ISIC). Table IX and Figure 5 present the results.

Rationale for ruling out SC1–SC4:

1) **SC1 & SC3 are not supported:** Synapse exhibits the lowest foreground coverage (7.7%) and lowest spatial entropy, yet possesses the highest GCS. Large target size or uniform dispersion does not necessitate large attention windows.
2) **SC4 is not supported (saturates):** In pixel space, even single polyps or skin lesions span across multiple 28×28 windows (WCR ≈ 1.0), demonstrating that component-level crossing is ubiquitous and cannot differentiate GCS tiers.
3) **SC2 is not supported on five datasets:** Boundary irregularity correlated with GCS on the initial 3 datasets (ρ=+0.50) due to CT image sharpness. Adding cardiac MRI (ACDC) and colonoscopy (CVC) drops this corre-

TABLE IX ASSOCIATION OF CANDIDATE FACTORS WITH EMPIRICAL GCS (M=28, FIVE BENCHMARKS).

Hypothesis	Metric	Synapse	ACDC	Kvasir	ISIC	CVC	ρ	Verdict	Rationale
SC1	Foreground ratio	0.077	0.040	0.152	0.214	0.094	-0.80	**Not supported**	Negative trend
SC2	Boundary complexity	1.355	0.928	1.023	1.261	1.163	$+0.00^{\dagger}$	**Not supported**	Zero correlation (5 datasets)
SC3	Spatial entropy	0.497	0.359	0.566	0.605	0.482	-0.50	**Not supported**	Negative trend
SC4	WCR (M=28 px)	0.999	0.954	1.000	0.999	1.000	-0.37	**Not supported**	Saturates near 1.0
SC5	Inter-class separation	**0.9996**	**0.547**	0.000	0.000	0.000	**+0.89**	**Strongest correlate**	Highest $\lvert\rho\rvert$ among candidates
Proxy	Mean fg. classes/slice	3.99	≈3.00	1.00	1.00	1.00	$+0.87^{\ddagger}$	**Insufficient**	ACDC is a counterexample

†Spurious partial correlation on 3 datasets ($\rho = +0.50$) collapsed to ρ=0.00 with 5 datasets. ‡Mean observed foreground classes per slice/image; insufficient as a sole explanation because ACDC has multi-class anatomy yet low GCS.

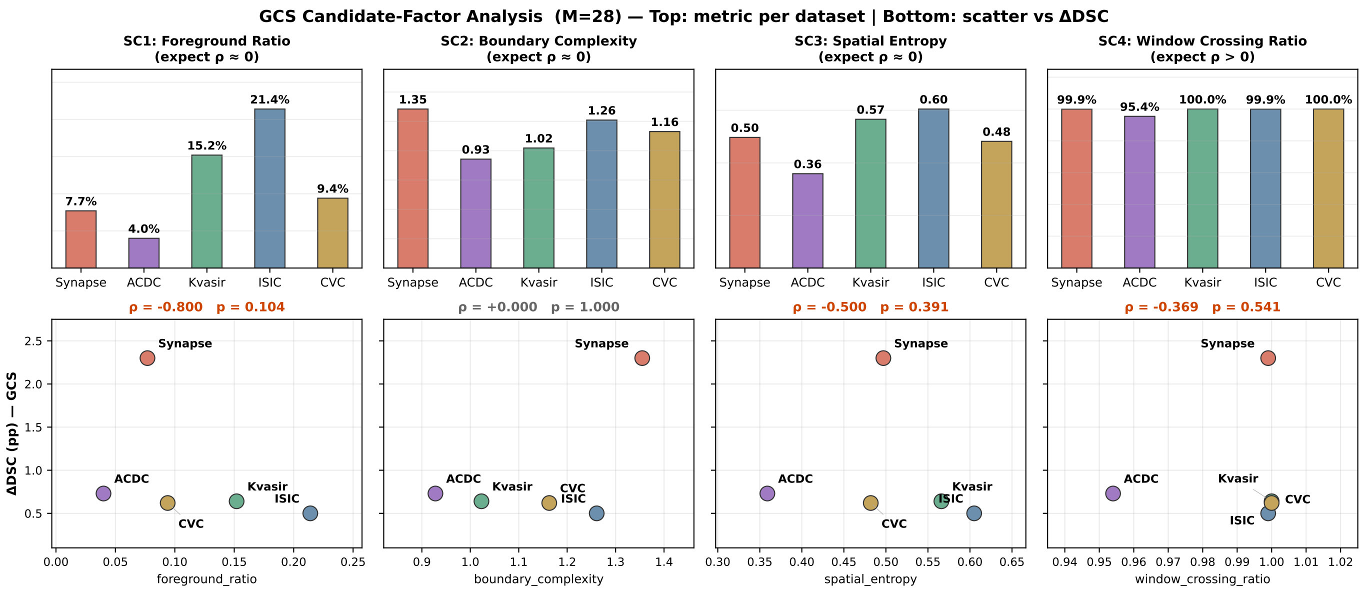

Fig. 5. Candidate-factor analysis across five medical segmentation benchmarks (SC1–SC4; SC5 is summarized separately in Table IX). Top row: each metric per dataset. Bottom row: the same metric plotted against ΔDSC, with Spearman ρ and corresponding p-value annotated. ACDC (violet) serves as a key discriminating benchmark: it shares the multi-class nature of Synapse but exhibits low context sensitivity ($\Delta\text{DSC} = 0.73\,\text{pp}$), inconsistent with SC1–SC4.

lation to ρ=0.00, consistent with a modality confound rather than a genuine effect.

C. Semantic Spatial Dependency (SSD): The Most Consistent Candidate Mechanism

Among all evaluated candidates, **SC5 (Inter-class Window Separation)** achieves the strongest rank correlation ($\rho = +0.89$) across all five datasets. This provides evidence that GCS is associated with **Semantic Spatial Dependency (SSD)**: *the degree to which correctly segmenting a target structure may require identifying its spatial relationship relative to other distinct semantic classes.*

$$\underbrace{\text{Inter-class SSD}}_{\text{Intrinsic task property}} \longrightarrow \underbrace{\text{Cross-structure spatial attention}}_{\text{Representational requirement}} \longrightarrow \underbrace{\text{High window sensitivity}}_{\text{Empirical GCS}}.$$

ACDC as a key discriminating case. To distinguish SSD from trivial class count (n_{classes}) as an explanation, the ACDC benchmark serves as a useful discriminating case. ACDC contains 4 semantic classes (background, RV, Myo, LV). If class count alone were the driver, ACDC would be expected to exhibit high GCS comparable to Synapse. Instead, ACDC exhibits $\Delta\text{DSC} = 0.73\,\text{pp}$, clustering tightly with binary tasks (0.50–0.64 pp), which is inconsistent with class count as a sole explanation.

SC5 is consistent with this pattern: cardiac structures are concentric and co-located within the cardiac silhouette (SC5 $=$ 0.547), so local windows capture sufficient multi-class context. In contrast, Synapse organs are dispersed across the entire abdominal cavity (SC5 $=$ 0.9996), where windowed attention at M=7 cannot attend to neighboring reference organs. Near-complete inter-class spatial separation was observed only for the high-GCS dataset in our benchmark set.

Within-dataset organ-level observation. As shown in Table X, organ-level sensitivities within Synapse are broadly consistent with an anatomical-SSD interpretation, though this per-organ breakdown is a supporting observation rather than an independently validated analysis (no separate per-organ SSD proxy was computed):

- **Higher-sensitivity structures:** The gallbladder ($\Delta\text{DSC} = 6.45\,\text{pp}$) is small and variable, located

TABLE X Per-organ DSC (%) and window sensitivity (ΔDSC) on Synapse (gate=pam, r=32, n=4 window sizes). Sorted by ΔDSC descending.

Organ	M=7	M=28	M=56	M=112	ΔDSC
Gallbladder	65.2	68.6	62.2	64.9	**6.45 pp**
Stomach	73.8	78.7	75.5	77.0	**4.97 pp**
Kidney (R)	78.1	82.2	82.1	80.5	**4.16 pp**
Spleen	87.5	89.4	85.9	86.4	3.52 pp
Pancreas	60.6	62.7	61.6	62.7	2.05 pp†
Kidney (L)	82.5	84.5	84.4	82.5	2.02 pp
Aorta	87.6	88.0	86.1	87.8	1.89 pp
Liver	93.7	93.3	93.4	93.6	**0.39 pp**
Mean	78.6	80.9	78.9	79.4	2.30 pp

†Pancreas scores 60–63% across all windows, reflecting a general segmentation ceiling rather than low SSD.

at the liver fossa; its identification plausibly benefits from liver-gallbladder spatial context. The stomach (ΔDSC $= 4.97$ pp) varies drastically in fill state, which may require adjacent organ boundaries for localization.

- **Lower-sensitivity structures:** The liver (ΔDSC $=$ 0.39 pp) is the dominant anatomical landmark recognizable from local parenchymal texture alone. The aorta (ΔDSC $=$ 1.89 pp) follows a rigid vertical cylindrical trajectory, rendering it largely window-robust.

D. Spatial Attention Allocation and Locality Prior

Why does windowed attention outperform global attention on Kvasir-SEG? Figure 6 visualizes the spatial attention distribution ($\|\text{output} - \text{input}\|_2$) for windowed ($M$=7) vs. global ($M$=112) PAM on Kvasir-SEG polyp images.

Quantitative measurement across 10 test samples confirms that windowed attention concentrates **62.3%** of its energy on the polyp region, compared to only **34.2%** for global attention (1.82$\times$ concentration ratio, consistent across 9/10 cases). In low-SSD tasks, global attention disperses representational capacity into irrelevant background parenchyma; constraining attention to local windows acts as an inductive bias that sharpens boundary delineation and filters spurious distant interactions.

VI. Optimization Dynamics and Routing Mechanics

Having examined how Semantic Spatial Dependency is associated with segmentation accuracy across tasks, we now examine the microscopic behavior of ApproxDA-TransUNet: how windowed approximation alters late-stage training dynamics (Section VI-A), and why learnable dual-branch attention gates settle near uniform routing (Section VI-B).

A. Windowed Attention and Late-Stage Training Variability

To investigate how spatial approximation affects the training landscape, we analyze optimization trajectories across representative datasets spanning the GCS spectrum (Synapse for high GCS; Kvasir-SEG and ISIC 2018 for low GCS). We define *validation DSC volatility* as the standard deviation of

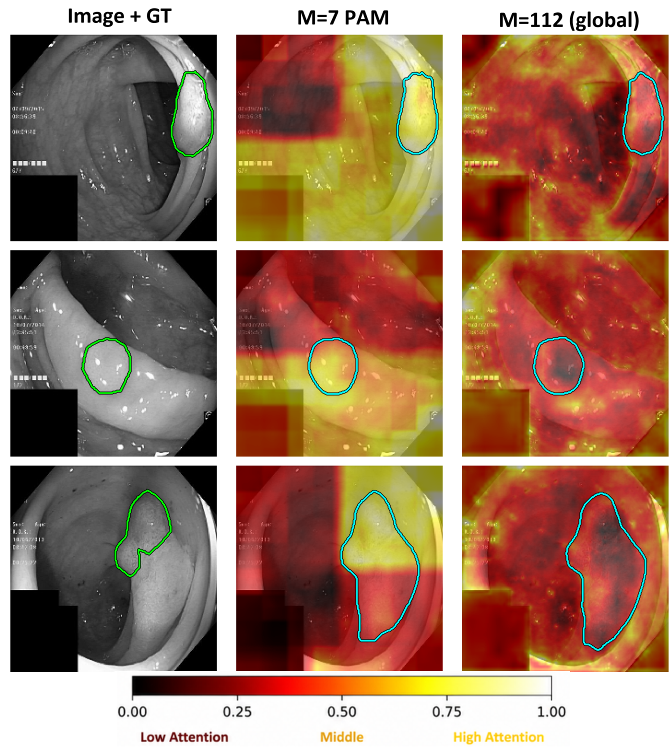

Fig. 6. PAM contribution maps ($\|\text{output} - \text{input}\|_2$, normalized per-panel) on Kvasir-SEG polyp images. Windowed attention (M=7) concentrates on polyp boundaries, whereas global attention (M=112) disperses into surrounding tissue.

validation DSC evaluated at 15-epoch intervals over the final 30% of training (epochs 210–300, 7 evaluation checkpoints). Lower volatility indicates less temporal variation in validation DSC during this measured late-training interval. Note that the ApproxDA configurations compared here use gate=pam for Synapse but gate=learn for Kvasir-SEG and ISIC 2018 (matching each dataset's best-performing configuration from Section IV); the reported cross-dataset differences may therefore reflect window size, gate mode (including gate collapse, Section VI-B), or both, rather than window size in isolation.

Table XI summarizes best validation DSC, test DSC, generalization gap (Val $-$ Test), and late-stage volatility for DA-TransUNet and ApproxDA-TransUNet.

Cross-task differences in late-stage validation variability. As illustrated in Figure 7, tracking validation DSC and training loss reveals a task-dependent pattern:

- **Low-GCS tasks:** The selected ApproxDA configurations exhibit substantially lower late-stage validation variability than DA-TransUNet on Kvasir-SEG (18.2$\times$ lower, 0.042 vs. 0.769) and ISIC 2018 (9.0$\times$ lower, 0.053 vs. 0.479).
- **High-GCS Synapse:** The opposite pattern is observed: ApproxDA-TransUNet exhibits 1.3$\times$ higher volatility than DA-TransUNet (0.542 vs. 0.409).

This pattern is directionally consistent with a locality-related hypothesis, but we note that the compared configurations differ in gate mode as well as window size (see caveat above), so window size, gate mode, and their interaction cannot be disentangled from this comparison alone. Controlled experiments holding gate mode fixed across datasets, together with

TABLE XI Optimization Volatility and Generalization Gap across Representative Benchmarks Spanning the GCS Spectrum. Volatility is measured as the standard deviation of validation DSC over epochs 210–300 (lower is more stable).

Dataset	GCS Tier	Model Architecture	Best Val (%)	Test DSC (%)	Val−Test Gap	Volatility (σ_{val})
Synapse	High	DA-TransUNet	79.52	79.80	−0.28	0.409
Synapse	High	ApproxDA (M=28)	81.02	80.94	+0.08	0.542
Kvasir-SEG	Low	DA-TransUNet	88.38	88.44	−0.06	0.769
Kvasir-SEG	Low	ApproxDA (M=7)	89.23	89.24	−0.01	**0.042** (18.2× reduction)
ISIC 2018	Low	DA-TransUNet	87.71	88.43	−0.72	0.479
ISIC 2018	Low	ApproxDA (M=7)	89.56	89.58	−0.02	**0.053** (9.0× reduction)

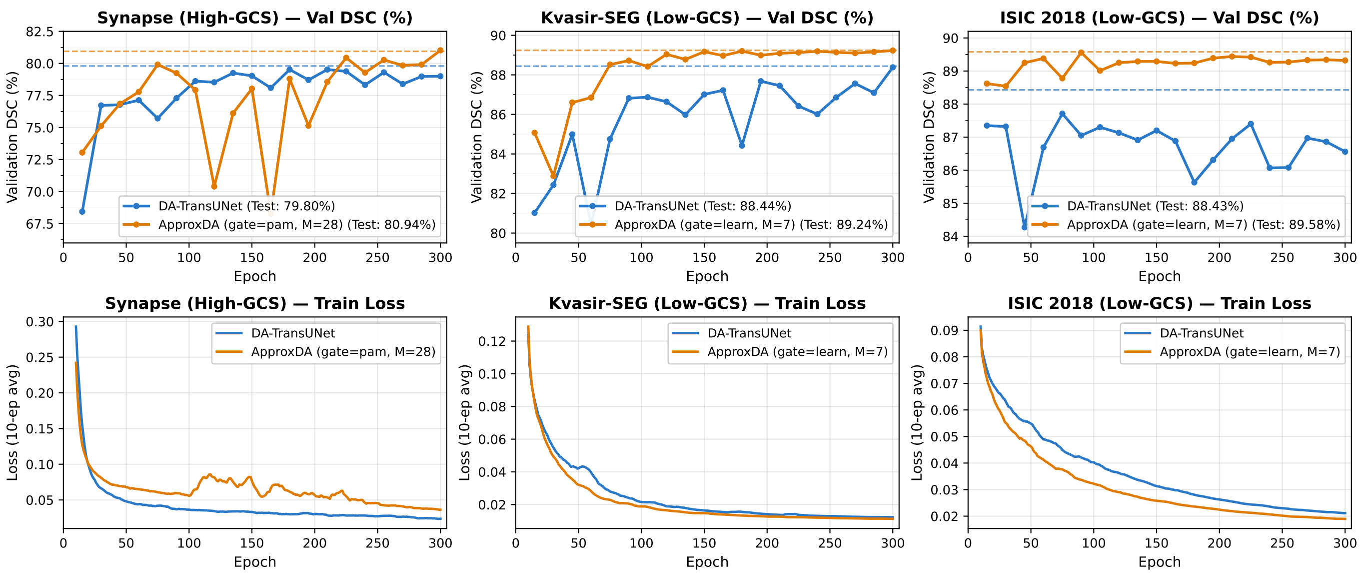

Fig. 7. Optimization trajectories for DA-TransUNet (blue) vs. ApproxDA-TransUNet (orange). Top row: Validation DSC trajectories across epochs (dashed lines indicate final test DSC). Bottom row: Corresponding training loss trajectories (10-epoch rolling averages).

direct gradient measurements, would be needed to isolate the mechanism underlying these observations.

Furthermore, across all evaluated settings, the generalization gap remains tight ($\leq 0.72\%$), with test performance closely tracking validation scores, suggesting the observed variability differences are not primarily an artifact of overfitting.

B. Gate Dynamics and Gradient Symmetry Collapse

We next investigate the routing dynamics between spatial (PAM) and channel (CAM) attention branches. While learnable scalar gates are widely utilized in dual-branch architectures to adaptively balance representations, empirical evaluations show that learned gating frequently underperforms fixed routing.

Table XII compares gate configurations on Synapse, including a robustness check at M=14, r=64 to verify gate collapse is not specific to the default M=7, r=32 setting. Fixed spatial attention (gate=pam, 78.64% DSC) consistently outperforms CAM-only (78.26%) and both learnable gate configurations (77.78% at M=7, r=32; 78.04% at M=14, r=64).

Mathematical derivation of gate collapse. Given the fused feature representation $\mathbf{F} = g\mathbf{F}_{\text{P}} + (1-g)\mathbf{F}_{\text{C}}$, the analytic gradient with respect to gate parameter g is:

$$\frac{\partial \mathcal{L}}{\partial g} = \frac{\partial \mathcal{L}}{\partial \mathbf{F}} \cdot (\mathbf{F}_{\text{P}} - \mathbf{F}_{\text{C}}) \,. \tag{7}$$

TABLE XII Gate Configurations and Gate-Collapse Robustness Check on Synapse

Gate Mode	g	DSC (%) ↑	HD95 (mm) ↓
gate=pam (M=7, r=32)	1.0	**78.64**	31.09
gate=cam (M=7, r=32)	0.0	<u>78.26</u>	<u>30.59</u>
gate=learn (M=14, r=64)	≈0.5*	78.04	**29.09**
gate=learn (M=7, r=32)	≈0.5*	77.78	34.29

*Gate collapsed to $g{\approx}0.5$ throughout training. The M=14, r=64 row uses a different window/rank setting than the other rows, included specifically to check that gate collapse is not an artifact of the default M=7, r=32 configuration.

At standard initialization ($g \approx 0.5$), the backpropagated gradients into both PAM and CAM branches are symmetric:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{F}_{\text{P}}} = g\frac{\partial \mathcal{L}}{\partial \mathbf{F}} \approx \frac{1}{2}\frac{\partial \mathcal{L}}{\partial \mathbf{F}}, \qquad \frac{\partial \mathcal{L}}{\partial \mathbf{F}_{\text{C}}} = (1-g)\frac{\partial \mathcal{L}}{\partial \mathbf{F}} \approx \frac{1}{2}\frac{\partial \mathcal{L}}{\partial \mathbf{F}}. \tag{8}$$

Under the assumption that branch representations become similar ($\mathbf{F}_{\text{P}} \approx \mathbf{F}_{\text{C}}$), the differential term $(\mathbf{F}_{\text{P}} - \mathbf{F}_{\text{C}})$ approaches zero, causing $\partial\mathcal{L}/\partial g \approx 0$. This analysis identifies a potential fixed-point tendency: once branches produce similar outputs, the gate gradient weakens, suggesting a tendency toward $g \approx 0.5$. Note that this reasoning concerns $\partial\mathcal{L}/\partial g$ and does not account for the full dynamics of the gate network parameters

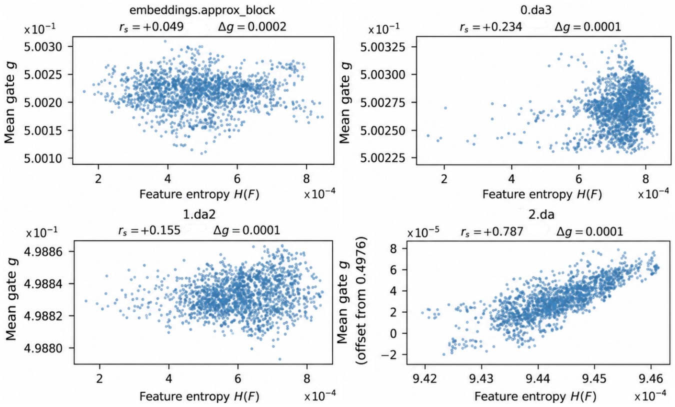

Fig. 8. Learned gate value g vs. input feature entropy $H(F)$ across four network stages (M=7, gate=learn). Gate values cluster tightly at $g \approx 0.5$ ($\Delta g \leq 2\times10^{-4}$ at every stage). Spearman rank correlation between entropy and g varies by stage (r_s=0.049, 0.234, 0.155, 0.787), including one stage with a fairly strong correlation; nonetheless, the absolute magnitude of gate variation remains negligible ($\leq 2\times10^{-4}$) at every stage.

W_g, sigmoid nonlinearity, or branch parameter updates, which may interact in complex ways.

Empirical verification across stages. As depicted in Figure 8, tracking learned gate values against feature entropy across four network stages reveals that gate values stay pinned at $g \approx 0.5$, with absolute variation $\Delta g \leq 2\times10^{-4}$ at every stage. The Spearman rank correlation between entropy and gate value varies across stages (r_s=$0.049, 0.234, 0.155, 0.787$), including one stage with a fairly strong correlation. Thus, while the gate value is not strictly independent of entropy at every stage, its practical range of variation is too small at any stage to provide meaningful adaptive routing.

These findings show that scalar gating failed to provide adaptive routing in the evaluated ApproxDA configurations. In this setting, fixed, task-aligned routing (e.g., gate=pam) provided a more reliable alternative; explicit structural asymmetries or diversity-promoting objectives are plausible directions for recovering adaptive routing in symmetric dual-branch architectures more generally.

VII. Conclusion and Discussion

This paper presented **ApproxDA-TransUNet**, an efficient, modular, and interpretable dual-attention framework for biomedical image segmentation. Across five heterogeneous clinical benchmarks spanning multi-organ CT, cardiac MRI, polyp endoscopy, colonoscopy, and dermoscopy, ApproxDA-TransUNet consistently matches or surpasses dense full-attention baselines while providing native DDP compatibility and shorter wall-clock training time in our two-GPU evaluation setting.

Beyond raw segmentation performance, this study systematically investigated the foundational principles governing attention approximation in medical imaging, yielding four interconnected insights:

1) **Task-Aligned Approximation as an Inductive Bias:** Attention approximation is more than a computational compromise to alleviate quadratic complexity. On low-GCS tasks, task-aligned approximation—whether through spatial restriction, low-rank projection, or both—can improve segmentation fidelity, plausibly by filtering out spurious long-range background dependencies. This benefit is not uniformly tied to compact locality: our best CVC-ClinicDB configuration uses global spatial scope (M=112) with only low-rank approximation, while Kvasir-SEG and ISIC 2018 benefit most from restricted windows (M=56 and M=7 respectively).
2) **Semantic Spatial Dependency (SSD) as the Strongest Explanatory Factor:** Through systematic candidate-factor analysis, we found that inter-class spatial window separation (SC5, ρ=+0.89) is the candidate most consistently associated with Global Context Sensitivity (GCS) across the five evaluated benchmarks. The ACDC benchmark serves as an important discriminating case: despite sharing multi-class anatomy with Synapse, its concentric cardiac structures yield low GCS, consistent with the SSD interpretation. Alternative candidates (SC1–SC4) are more weakly correlated or exhibit negative trends across the full benchmark set.
3) **Cross-Task Differences in Late-Stage Training Variability:** By tracking validation dynamics across epochs, we observed lower late-stage validation variability for the selected ApproxDA configurations on low-GCS tasks Kvasir-SEG and ISIC 2018 (standard deviation reduced by 9.0× to 18.2× relative to DA-TransUNet), while the opposite pattern is observed on high-GCS Synapse. Because the compared configurations differ in gate mode as well as window size across datasets, this pattern is directionally consistent with a locality-related hypothesis but does not isolate window size as the sole cause.
4) **Gradient Symmetry and Gate Collapse Dynamics:** We provided a gradient-based fixed-point analysis showing that, under the assumption that branch outputs become similar, the gate gradient vanishes—suggesting a tendency toward near-uniform blending ($g \approx 0.5$). Empirical measurements confirm gate values pinned at g=0.5000 $\pm$ 0.0002 in our ApproxDA configuration. Consequently, fixed spatial attention (gate=pam) provides a more reliable inductive prior in this setting.

A. Design Considerations for Medical Vision Practitioners

Based on our findings, we suggest the following exploratory heuristics for designing attention architectures in biomedical segmentation. These are derived from five benchmarks and should be treated as hypotheses for further validation rather than universal decision rules:

- **Pre-training SC5 Assessment:** Practitioners can compute inter-class window separation (SC5) directly on annotation masks prior to model training as a lightweight diagnostic.
  - *Tasks with high inter-class spatial separation (e.g., SC5 ≈ 1.0 in Multi-Organ CT):* Tend to exhibit higher window sensitivity. Multi-scale window evaluation ($M \in \{7, 28, 56\}$) is advisable.

– *Tasks where all classes share a single spatial region (SC5 $= 0$ for binary datasets, SC5 < 0.6 for cardiac MRI in our study):* Tend to exhibit broad window robustness. Compact localized windows ($M{=}7$ or $M{=}28$) are effective starting points.

These thresholds are exploratory heuristics derived from the five benchmarks evaluated here; held-out prospective validation on additional datasets remains future work.

- **Branch Fusion Design:** Avoid unconstrained learnable scalar gating across symmetric spatial and channel branches unless explicit architectural asymmetries or regularizing priors are introduced. Fixed spatial routing (gate=pam) provides more reliable feature refinement in our evaluated configurations.

B. Limitations and Future Horizons

All window-size and GCS results in this paper are reported from single training runs per configuration; the low-GCS spans in particular (0.50–0.73 pp for ACDC, Kvasir-SEG, CVC-ClinicDB, and ISIC 2018) are small enough that their distinguishability from training-seed variance has not been directly measured. Multi-seed replication of the highest- and lowest-sensitivity configurations is left as important future validation. Three further promising avenues for future inquiry emerge from this investigation:

1) **Intra-Sample Context Heterogeneity:** While ACDC is categorized as low-GCS overall (ΔDSC $= 0.73$ pp), ApproxDA-TransUNet achieves the strongest RV result among compared methods, remains competitive with DA-TransUNet on Myo (85.91% vs. 85.98%), and improves LV over DA-TransUNet (91.40% vs. 90.67%) although global Transformer baselines retain an advantage on LV. Future architectures could incorporate dynamic multi-window partitioning within the same block to dynamically allocate receptive fields according to substructure context requirements.
2) **Adaptive Low-Rank Dimension Allocation:** In this work, representation rank $r{=}32$ was held constant. Content-adaptive rank allocation could dynamically assign higher projection fidelity to complex, high-frequency boundary regions.
3) **Direct Geometry-to-Hyperparameter Synthesis:** While SC5 is associated with the observed GCS tiers in our benchmark set, it cannot characterize within-class global dependencies in binary segmentation tasks (Section V-B). Directly synthesizing optimal continuous window hyperparameters M^* from training mask topological descriptors remains an exciting open research challenge.

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